Online Supplementary Material for "Reducing, Holding and Expanding the Size of the State: A Modular Stochastic Framework for Evaluating Fiscal Reform Scenarios in Brazil, with an Application to the 2026 Presidential Programmes"

Flaviano A. Moreira*

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This document collects, in full, the extended literature review and the off-by-default diagnostic extensions summarised in the main manuscript. Section and table labels here are internal to this supplementary document; the main manuscript refers to this material descriptively rather than by number, since the two documents are typeset and paginated separately.

A Extended literature review

This appendix gives the full discussion summarised in the main manuscript's literature-review section, moving from the broad macro-fiscal debates that motivate the model's overall design to the narrow, often Brazil-specific estimates that calibrate individual blocks. Four of the citations below (Levy Yeyati and Sturzenegger, 2023; Lago-Peñas, Cadaval-Sampedro and Herrero-Alcalde; Hanousek and Kočenda, 2011; Balladares and García-Miralles, 2026) were themselves published in *Fiscal Studies*, one of several applied public-finance journals whose recent work bears directly on this paper's own sub-literatures.

A.1 The composition of fiscal consolidation

A long empirical literature asks whether fiscal consolidations differ in their output cost according to whether they are achieved through spending restraint or revenue increases. Alesina and Ardagna (2010) and the subsequent plan-based approach of Alesina et al. (2015,?) find that multi-year adjustment plans built mainly from spending cuts produce markedly milder and shorter output losses than plans built mainly from tax increases, a gap concentrated in private investment rather than consumption, and robust to whether the economy enters the adjustment in recession (Alesina et al., 2017). This finding is contested: Guajardo (Leigh and Pescatori) use Blanchard–Perotti-style narrative identification of exogenous consolidation shocks to reject the Alesina–Ardagna "expansionary austerity" hypothesis more broadly, finding both spending- and tax-based consolidations contractionary once endogenous and anticipated fiscal actions are stripped out, with any residual composition asymmetry a secondary rather than central finding

*Independent research; no institutional affiliation. Correspondence: flaviano.a.moreira@gmail.com. ORCID: 0009-0009-0783-5477.

of their design; Perotti (2013) and Jordà (Schularick and Taylor) likewise caution that composition effects in the Alesina–Favero–Giavazzi tradition may partly reflect omitted monetary accommodation and the specific European sample. The present model does not take a position in this debate: it inherits whatever composition the user specifies, pension indexation, subsidies, administrative efficiency, public investment, or means-tested transfers, and reports the growth, debt and distributional consequences of that composition using Brazil-specific multipliers, rather than importing a foreign average effect.

A.2 Fiscal multipliers: size, state-dependence and emerging-market evidence

A logically prior question to the composition debate is how large a fiscal multiplier is in the first place, and whether it varies with the state of the economy. Auerbach and Gorodnichenko (2012) develop the regime-switching methodology that established multipliers are substantially larger in recessions than expansions in advanced economies. Ramey and Zubairy (2018) partially qualify that result on a longer US historical sample, finding multipliers below unity regardless of the degree of slack once defence-spending identification and a longer sample are used, while Ramey (2011, 2019) survey the resulting range, broadly 0.8 to 1.5, and the post-2008 renaissance in multiplier estimation more generally. The closest cross-country precedent to this model's own package-specific approach is Ilzetzki (Mendoza and Végh), who show multipliers vary systematically with exchange-rate regime, trade openness, public debt and the level of development, with emerging, relatively closed, high-debt economies such as Brazil tending towards the lower end of the range. European growth-forecast errors during 2010–2011 austerity, Blanchard and Leigh (2013) show, are consistent with multipliers considerably larger than the then-standard 0.5 assumption, a finding with obvious relevance to the persistent-versus-transitory multiplier distinction drawn throughout this paper; Nakamura and Steinsson (2014) supply the leading regional-multiplier identification strategy as a cross-check on these aggregate time-series estimates. Two IMF working papers estimate multipliers specifically for Latin American and Caribbean economies: Carrière-Swallow (David and Leigh) and Restrepo (2020) both find multipliers below one, with Restrepo's country-by-country estimates also revealing substantial heterogeneity even within the region, a caution against assuming Brazil's multiplier resembles its neighbours' any more closely than it resembles an advanced-economy average. Born (Müller and Pfeifer) connect multiplier state-dependence to the sovereign-risk-premium channel used in the main manuscript's discussion of the sovereign-risk-premium channel, showing that whether austerity narrows or widens spreads depends on the fiscal starting position. Two Brazil-specific estimates beyond Sanches (2026) and Carvalho and Sanches (2019) corroborate the order of magnitude used here: Pires (2014) estimates a Markov-switching, regime-dependent multiplier for Brazil using IPEA-affiliated methods, and Resende and Pires (2021) find a low aggregate multiplier but markedly higher public-investment and targeted-transfer multipliers, consistent with this model's category-specific rather than single-number approach.

A.3 Stochastic and probabilistic debt-sustainability analysis

The deterministic debt-sustainability exercises conducted by the IMF and finance ministries have long been criticised for reporting a single central path with no account of the uncertainty around growth, interest rates and the primary balance (IMF, 2003). Celasun (Debrun and Ostry) proposed replacing the deterministic path with a stochastic simulation, in which a vector autoregression on the macro-fiscal block generates a joint distribution of shocks that is then propagated through the debt-accumulation identity to produce fan charts of future debt ratios. An important direct antecedent applies closely related risk-management machinery to Brazilian sovereign debt specifically: Garcia and Rigobon (2004) show that a debt path

can look sustainable at its central projection while still carrying a non-trivial probability of an unsustainable path once the full joint distribution of shocks is considered, precisely the distinction this paper's own Tier 1/Tier 2 gate (the main manuscript's back-testing section) is built to respect. Karam and Hostland (2005, 2006) and Mendoza and Oviedo (2004), the latter applying a probabilistic fiscal-solvency framework to Brazil, Colombia, Costa Rica and Mexico directly, are the other principal stochastic-DSA antecedents. This approach, and its descendants in the IMF's Debt Sustainability Framework (IMF, 2013) and Sovereign Risk and Debt Sustainability Framework (IMF, 2022), is the direct ancestor of the Monte Carlo design used here. Closer to the present, Levy Yeyati and Sturzenegger (2023) reformulate the same underlying question as a balance-sheet solvency exercise rather than a flow-based fan chart, valuing a government's assets and liabilities directly; the two approaches are complementary rather than competing, and a balance-sheet cross-check of this paper's own debt projections is a natural extension we have not undertaken.

More recent work has moved from fan charts around a single mean path towards full conditional-distribution estimation. Prasad et al. (2019) introduce growth-at-risk quantile methodology into IMF country surveillance; Lian (Presbitero and Wiriadinata) extend it to the interest-growth differential itself, and Furceri et al. (2025) extend it further into an explicit "debt-at-risk" quantile framework. This is the natural next step for a model such as this one, which currently reports percentiles of a resampled distribution rather than a true conditional quantile regression (a limitation the main manuscript's discussion section returns to). Kaffo Melou (Sumlinski and Geiregat) show both the portability and the limits of the fan-chart approach by applying it outside its original emerging-market sample, to post-HIPC low-income countries, a useful caution against assuming any stochastic-DSA machinery transfers mechanically across very different debt and institutional contexts. Our implementation differs from all of the above by sampling structural parameters (multipliers, elasticities, growth-accounting inputs) directly from triangular or uniform ranges drawn from named studies, rather than estimating a reduced-form shock covariance matrix; this trades statistical efficiency for interpretability, since every sampled draw can be traced back to a specific coefficient and its source. Bohn (1998) provides the foundational fiscal-reaction-function literature against which any debt-sustainability exercise should eventually be checked. Section A.11 reports what that check looks like for Brazil specifically, and Appendix B reports the diagnostic this model now runs against it.

A.4 Debt thresholds, nonlinearity and the interest-growth differential

Reinhart and Rogoff (2010) and its contested 90%-of-GDP growth-threshold claim, together with the spreadsheet-error and sample-selection correction of Herndon (Ash and Pollin), is the cautionary precedent this paper has in mind whenever it is tempted to summarise a debt path with a single threshold rather than a distribution. Égert (2015) shows why directly: the apparent threshold, where it is detectable at all, is highly sensitive to country and period coverage and typically falls well below 90% once outliers and reverse causality are handled. Pescatori (Sandri and Simon) find no evidence of any specific debt level beyond which medium-term growth collapses, arguing instead that the debt *trajectory*, rising or falling, matters more than the level; Panizza and Presbitero (2014) go further still, using an instrumental-variables strategy built on valuation effects to argue that the negative debt–growth correlation largely disappears once reverse causality is properly addressed, so that no robust causal effect of debt on growth survives. Chudik et al. (2017) reach a related conclusion using dynamic heterogeneous-panel methods across forty countries: no single threshold applies across countries, but a high-debt country on a falling trajectory can grow as fast as a lower-debt peer. Woo and Kumar (2015),

by contrast, find a more conventional, if modest, negative panel association operating mainly through slower productivity growth. For Brazil specifically, the only threshold-autoregressive estimate of a Brazilian fiscal reaction trigger we are aware of is Simonassi and Lima (2005), who find that stabilisation is activated only once the deficit crosses an estimated threshold near 1.7% of GDP. That is a different quantity from the debt-level thresholds discussed above, but it is the closest Brazil-specific evidence that fiscal responses themselves may be nonlinear, and it bears directly on the nonlinear risk-premium diagnostic introduced in Appendix B.

A separate literature, arguably more consequential for a country with Brazil's interest-growth gap, asks not whether a debt *level* is dangerous but whether the interest-growth differential itself is structurally favourable or unfavourable. In his 2019 AEA presidential address, Blanchard (2019) argues that historically $r < g$ has been closer to the norm than the exception for the United States, so that debt rollover under such conditions carries limited welfare cost. Furman and Summers (2020) propose real interest payments relative to GDP, rather than the debt ratio itself, as the more informative fiscal-space metric, precisely because it is this differential, not the stock, that drives accumulation; Rachel and Summers (2019) trace the decline in neutral real rates to deeper secular-stagnation forces. Mian (Straub and Sufi,S) formalise how indebted demand depresses the natural rate and, in a model with a zero lower bound, can make the relationship between the interest-growth gap and sustainable fiscal space considerably more complicated than a simple threshold; we take this as a defensible extrapolation of their secular-stagnation argument rather than a result they estimate directly. This literature directly motivates the finding, reported in the main manuscript's headline-results section, that the interest-growth differential rather than the primary balance is the dominant source of Brazil's own projected status-quo debt accumulation.

A.5 Distributional and social consequences of adjustment

Ball et al. (2013) use a sample of advanced-economy consolidation episodes identified through the Devries et al. narrative approach to show that fiscal consolidation raises the Gini coefficient, lowers the wage share and raises long-term unemployment, with effects that persist for several years after the initial shock. Woo (Bova) extend this to a wider advanced- and emerging-market panel plus country case studies, confirming that spending-based consolidations tend to be more regressive than tax-based ones, though progressive design (protecting targeted transfers, as the present model's default package composition does) can materially offset the distributional cost. Agnello and Sousa (2014, 2012) reach a broadly similar conclusion for eighteen industrialised economies over three decades; Mulas-Granados (2005, 2006) is the earlier European-adjustment-episode precedent on which much of this literature, including Ball et al. (2013), draws for its policy-trade-off framing. More recent work refines the dynamics. Using local-projection methods across seventeen OECD countries, Heimberger (2020) shows the Gini effect builds from around 0.4 points on impact to 0.6 points at medium horizons and is larger for spending-based and post-crisis consolidations, a time profile qualitatively similar to the one our own household microsimulation produces mechanically. Brinca et al. (2021) offer a useful complement running in the opposite causal direction, showing that *higher* initial income inequality itself amplifies the output cost of a given consolidation, a channel this model does not yet capture since its inequality-to-growth feedback (the main manuscript's model-architecture section) remains switched off pending a wired Gini path. Leventi and Matsaganis (2013) use the EUROMOD microsimulation model to separate the distributional effects of fiscal consolidation from those of the accompanying recession in Greece, a methodological strategy close in spirit to this paper's own comparative-statics design. A complementary political-economy perspective comes from Lago-Peñas (Cadaval-Sampedro and Herrero-Alcalde), who show in a survey of Spanish voters

that tolerance for austerity depends sharply on its financing instrument and the respondent's own ideology, a reminder that the distributional incidence this model computes mechanically is also, separately, a political variable the model does not attempt to capture. The elasticity our household microsimulation recovers for a comparable Brazilian package, a 0.9-point Gini increase at year eight for an approximately 1% of GDP consolidation, is of the same order of magnitude as this literature's central estimates, though it is produced mechanically from PNAD cash components and a labour-market pass-through rather than estimated from cross-country consolidation episodes, so the agreement is suggestive rather than a validation.

On the health side, Szklo (Clarke and Rocha) estimate the elasticity of infant mortality to public health spending around Brazil's EC 29/2000 constitutional amendment using a difference-in-differences design across municipalities, finding a central elasticity of −0.126, while Gallet and Doucouliagos (2017) provide a meta-regression elasticity of life expectancy to total health spending of 0.04. These Brazilian and cross-country estimates should be read against a standing scepticism: Filmer and Pritchett (1999) find that cross-national public health spending alone explains remarkably little of the variation in child mortality once income and female education are controlled for, which is why this model applies its elasticity only to Brazil's own within-country variation rather than extrapolating a cross-country average. Bokhari (Gai and Gottret) and Moreno-Serra and Smith (2012, 2015) find the mortality-reducing association of public health spending and coverage expansion is in fact stronger in developing- than in advanced-economy samples, at least broadly consistent with a Brazilian elasticity of the magnitude Szklo (Clarke and Rocha) estimate. The strongest Brazil-specific evidence that a targeted transfer programme causally reduces child mortality comes from Rasella et al. (2013), whose study of Bolsa Família finds effects concentrated in poverty- and malnutrition-related deaths, a useful complement to this model's health-spending channel since it operates through a different, income-side mechanism the household microsimulation only partially captures. Litschig and Morrison (2013); Litschig (2012) identify the effect of municipal spending on schooling outcomes in Brazil using a close-election and a royalty-windfall design respectively, and are the source of this model's schooling elasticity. Hanushek and Woessmann (2012) provide the leading causal, immigrant difference-in-differences evidence that schooling *quality*, not years of schooling alone, drives the human-capital-to-growth channel; Card (1999) remains the standard reference for the likely downward bias of ordinary-least-squares returns-to-schooling estimates relative to instrumental-variables ones. Finally, Acemoglu and Johnson (2007) and Bloom (Canning and Sevilla) offer opposing readings of the health-to-growth channel, the former finding a limited long-run output effect from large historical mortality declines and the latter a sizeable one from cross-country panel estimates, which is why the present model samples this coefficient from a wide uniform range rather than adopting either point estimate.

A.6 Household microsimulation and fiscal-incidence methodology

The household-microsimulation block's comparative-statics design follows a long methodological tradition. Bourguignon and Spadaro (2006) set out the case for static microsimulation as a transparent, auditable tool for evaluating redistribution policy, precisely because every household's exposure to a policy change can be traced, at the cost of foregoing any behavioural response; Immervoll and O'Donoghue (2009) and Figari (Paulus and Sutherland) extend this into the design and validation lessons of large multi-country tax-benefit microsimulation models such as EUROMOD (Sutherland and Figari, 2013). The closest methodological relative to this paper's distributional block among Latin-American-focused work is the Commitment to Equity (CEQ) project, summarised in Lustig (2018)'s handbook and applied to Brazil directly

by Higgins and Pereira (2014), who find that regressive consumption taxation offsets much of the redistributive gain from Brazil's direct taxes and transfers, a finding this model's own instrument-specific incidence variants (the main manuscript's model-architecture section) are built to let a user reproduce or contest rather than assume away. Higgins and Lustig (2016) introduce the complementary concept of "fiscal impoverishment": a progressive, inequality-reducing tax-and-transfer system can still leave some households worse off in absolute terms, a distinction this model's poverty-headcount outputs do not yet report separately from its Gini and top-share outputs, a gap worth closing in future work. Lustig (Pessino and Scott) frame the wider Latin American CEQ comparison within which the Brazilian application sits, and Bourguignon (Ferreira and Lustig) is the earlier World Bank volume applying microsimulation and decomposition methods to income-distribution dynamics across Brazil and its neighbours. Closest to the present in method, Balladares and García-Miralles (2026) use administrative tax-record microsimulation to quantify fiscal drag in Spain, a recent demonstration that static, auditable microsimulation of exactly this kind remains a live and publishable way to answer a current policy question, which is the standard this paper's own household block aspires to.

A.7 Sovereign risk premia and fiscal fatigue

The model's debt-dependent risk-premium channel (the main manuscript's model-architecture section) draws its central coefficient from Baldacci and Kumar (2010), but the wider literature on how debt and fiscal fundamentals feed through to sovereign borrowing costs is considerably richer. Edwards (1984, 1986) provide the foundational empirical link between debt and debt-service ratios and the spread charged on developing-country borrowing. A similar relationship holds for a modern panel using EMBI-type spreads: Akitoby and Stratmann (2008) find debt-financed spending raises spreads more than tax-financed spending of the same size, a composition effect this model's revenue-elasticity channel (the main manuscript's model-architecture section) is built to let the user explore. Ardagna (Caselli and Lane) find the debt effect on long-term rates in OECD economies is concentrated in periods when debt is *above* its own historical average, an empirical analogue, well short of a structural model, of a nonlinear or threshold-dependent premium. The explicitly nonlinear theoretical case is made most directly by two papers: Ghosh et al. (2013)'s "fiscal fatigue" framework models a primary-balance reaction function that flattens as debt rises, producing an endogenous fiscal limit, while Bi (2012) derives a stochastic fiscal limit from a dynamic Laffer curve along which risk premia rise nonlinearly once debt passes a country-specific threshold. These two papers are the direct theoretical basis for the nonlinear, fiscal-fatigue-style risk-premium diagnostic introduced in Appendix B. Reinhart (Rogoff and Savastano)'s concept of "debt intolerance", the finding that safe debt thresholds are far lower, and more history-dependent, for emerging markets than for advanced economies, further motivates treating any single premium coefficient as provisional rather than structural; Mendoza and Yue (2012) provide the leading general-equilibrium unification of sovereign default and the business cycle, a theoretical benchmark against which this model's reduced-form premium channel is a deliberately modest approximation. For Brazil and its emerging-market peers specifically, Gupta (Mati and Baldacci) find fiscal consolidation narrows spreads especially for countries with a default history, and Campos and Cysne (2021) estimate country-specific rather than common debt limits for eighteen emerging economies, rejecting a single threshold in a manner consistent with Égert (2015)'s growth-threshold finding above. Ornelas (2017) and Rocha and Moreira (2012), both Brazilian-institution-authored, provide the closest direct estimates of the debt-to-spread elasticity for Brazil and comparable emerging markets, broadly consistent with, if somewhat larger than, the Baldacci and Kumar (2010) coefficient this model currently uses as its central case.

A.8 Governance, corruption and public-spending leakage

A distinct but related literature asks how much of public spending is diverted before it reaches its intended beneficiaries, and how this interacts with fiscal policy. Ferraz (Finan and Moreira) use random federal audits of Brazilian municipal education transfers to estimate that corruption is found in roughly a third of audited municipalities, diverting around 8% of audited education resources on average. Using the same audit programme, Avis (Ferraz and Finan) show that a past audit visit reduces subsequently measured corruption by about 8% across nearly two thousand municipalities, evidence this model uses as a lower bound on how much a credible anti-corruption programme could remove. Litschig and Zamboni (2018) extend this audit literature with a randomised increase in audit risk across Brazilian municipalities and find procurement corruption falls by 10–15 percentage points, though health- and cash-transfer-programme compliance is unaffected, a useful caution that the leakage parameter's effectiveness may be programme-specific rather than uniform (a simplification the present model's single leakage share does not yet relax). Colonnelli and Prem (2022) and the companion firm/worker-level study of Colonnelli et al. (2022) use the same Brazilian federal audit programme, the Controladoria-Geral da União (CGU), to show audits raise firm entry and growth in government-dependent sectors while penalising politically connected firms, rejecting a "grease the wheels" reading of corruption. Using discontinuities in Brazilian federal transfers, Brollo et al. (2013) show that larger unearned fiscal windfalls themselves raise measured municipal corruption and lower the quality of elected candidates, a result directly relevant to any scenario, such as a commodity-revenue boom, in which this model's external block delivers an unusually large fiscal windfall. Bobonis (Cámara Fuertes and Schwabe) provide a contrasting Puerto Rican result: pre-election audits reduce corruption and raise reelection chances, but the deterrent effect does not persist into later, non-electoral audit rounds, a caution against assuming the Brazilian audit evidence this model relies on implies a permanent, rather than temporary, leakage reduction.

Mauro (1995) established the cross-country association between perceived corruption and lower investment and growth that motivates treating corruption as a growth-relevant variable at all. Mauro (1998) extends this, showing corruption systematically skews public spending away from education and towards more easily diverted categories, a composition effect related to, but distinct from, the leakage-share mechanism used here; Tanzi and Davoodi (1997) similarly find corruption raises the *volume* of public investment while lowering its productivity and the revenue available to finance it, a quantity-versus-quality distinction this model's single leakage parameter does not yet separate. Gründler and Potrafke (2019) provide a more recent panel estimate of the long-run output effect of an improvement in the Transparency International Corruption Perceptions Index, which we hold off by default pending verification of a cross-country standard-deviation input. Hanousek and Kočenda (2011) close the loop most directly for this paper's purposes, linking corruption and regulatory quality to public investment, deficits and debt across the new EU member states, and finding that lower corruption is associated with better fiscal outcomes on both margins, a structural pairing of governance and debt dynamics that this model keeps as two formally separate blocks (G1/G2 and the debt identity) rather than one estimated relationship. The benchmark for the difficulty of measuring leakage directly, rather than through perceptions, is Olken (2007)'s missing-expenditure approach on Indonesian road projects. A closely related finding is Reinikka and Svensson (2004)'s "local capture" result, that only around 13% of an Ugandan school capitation grant reached its intended schools on average, and Fisman and Svensson (2007), who show using Ugandan firm-level data that bribery has roughly three times the negative effect on firm growth of the statutory tax rate, supporting a "corruption as tax" framing. Golden and Picci (2005) propose inferring corruption objectively from the gap between reported infrastructure spending and the physical infrastructure actually

delivered, a methodological contrast to the survey-based perception indices this model explicitly avoids using as a model input (the main manuscript's model-architecture section). Our leakage parameter is calibrated on the comparable Brazilian audit evidence rather than attempting a similar missing-expenditure exercise, which the available Brazilian audit microdata does not support at the scale required.

A.9 Fiscal rules and the political economy of public debt

Both 2026 candidate programmes modelled in the main manuscript's scenario-design section are, formally, fiscal rules (an expenditure rule and a balance-targeting rule), and so inherit a large, separate literature on fiscal-rule design that the model's own machinery is agnostic to but that any reader evaluating the two programmes' relative merits should have in view. Kopits and Symansky (1998) provide the foundational IMF taxonomy of rule types, balanced-budget, debt, expenditure and revenue rules, and the credibility-versus-flexibility trade-off between them. Using a dataset spanning eighty-one countries, Schaechter et al. (2012) document that post-2008 rules increasingly combine several rule types with escape clauses, a description that fits Brazil's own layered combination of the constitutional spending ceiling (Complementary Law 200/2023) and the pre-existing "golden rule" (Article 167, assessed directly for Brazil by Pires, 2019) reasonably well. Cordes et al. (2015) is the closest empirical comparator to the incumbent's programme specifically, evaluating the compliance record and investment effects of thirty-three expenditure rules worldwide. The standard real-world evaluation of an operating expenditure-ceiling-type rule is Feld and Kirchgässner (2008)'s study of the Swiss debt brake, which finds it durably reduced deficits without the escape-clause erosion some critics predicted, a useful, if institutionally distant, comparator for assessing whether Brazil's own ceiling is likely to bind. López-Villavicencio and Zoumenou (2025) offer a timely, directly comparable recent finding for the European Union, that only compliance with a binding deficit or debt rule, not an expenditure rule considered alone, robustly improves subsequent fiscal outcomes, which is worth weighing against the Brazilian ceiling's own design. Debrun and Kumar (2007) and Wyplosz (2018) examine the complementary role of independent fiscal councils in making numerical rules credible, relevant to Brazil's own Independent Fiscal Institution (IFI), whose most recent technical assessment of the 2023 spending framework we draw on directly in the main manuscript's scenario-design section (Instituição Fiscal Independente, 2023); Debrun (Hauner and Kumar) survey the wider international evidence on independent fiscal agencies. At a more foundational level, Alesina and Tabellini (1990) and Alesina and Passalacqua (2016)'s handbook survey model why governments with divergent political preferences over-borrow relative to a social optimum, a channel that operates independently of whichever numerical rule is nominally in force, while von Hagen and Harden (1995) and Persson and Tabellini (1999) are the classic references connecting budget-process centralisation and electoral-system design to fiscal discipline more broadly. None of this literature is wired into the model; it is offered here as the frame within which a reader should situate the two 2026 programmes' chosen closure rules, which this paper deliberately declines to rank (the main manuscript's scenario-design section).

A.10 Stock-flow adjustments, quasi-fiscal operations and fiscal transparency

The model's single most consequential finding is that its debt-level back-test fails specifically because of discretionary, non-forecastable "stock-flow adjustment" operations (the main manuscript's back-testing section), and this sits within a literature that has independently documented the same phenomenon across many countries. Using 163 countries over three decades, Weber (2012) shows stock-flow adjustments are a large and persistent, not merely occasional,

source of debt accumulation worldwide, only partly explained by observable balance-sheet effects and contingent liabilities, and systematically smaller in more fiscally transparent countries, directly consistent with this paper's own finding that Brazil's residual bears no relationship to any observable macroeconomic covariate. Seiferling (2013) refines this using government balance-sheet data directly, arguing that complete financial reporting, not a transparency *score* per se, is what narrows the gap. The closest empirical analogue to this paper's own 3.9-percentage-point single-year residual is Jaramillo (Mulas-Granados and Kimani)'s examination of 179 historical debt-spike episodes: spikes are driven primarily by stock-flow adjustments rather than by primary deficits or weak growth, and a large stock-flow adjustment predicts debt subsequently *failing* to decline, a genuinely alarming precedent for interpreting Brazil's own recent experience. Andrian (Rodríguez and Valencia) extend this to a hazard-model framework across 172 countries, finding a one-percentage-point rise in the stock-flow adjustment raises the probability of a debt surge by roughly fifteen per cent. Afonso and Jalles (2019) show that stronger fiscal-rule frameworks moderate the stock-flow-adjustment channel, while Casalin (Cerniglia and Dia) show that in many countries stock-flow adjustments function as a hidden, debt-management-driven component of the true cost of debt service, a framing that fits the Brazilian Treasury-to-BNDES lending and Central Bank result-transfer operations this paper identifies as the dominant residual unusually well. The foundational IMF treatment of *why* such operations occur at all is Mackenzie and Stella (1996): public financial institutions and central banks routinely run "quasi-fiscal" activities that shift genuine fiscal costs off the conventional budget, a description that applies directly to BNDES's role in the Brazilian episode. Bova et al. (2016) assemble the first systematic cross-country dataset of realised contingent-liability costs, finding an average fiscal cost of around six per cent of GDP, far higher for financial-sector bailouts, that clusters in crises and is, like this paper's own SFA residual, poorly predicted by observable covariates. Alesina (Prati and Tabellini) provide the classic theoretical link between debt-management choices, such as maturity structure and market confidence, and self-fulfilling crisis risk. Finally, the European Commission's own recurring statistical reporting on member-state stock-flow adjustments (Eurostat, 2024) documents magnitudes of a similar order, up to two to three per cent of GDP, even among countries operating under a far more codified fiscal-transparency regime than Brazil's, suggesting the phenomenon this paper documents is a structural feature of sovereign debt accounting rather than a peculiarly Brazilian failing. No peer-reviewed economics-journal article appears to model Brazil's own 2015–2016 *"pedaladas fiscais"* episode specifically as a stock-flow-adjustment event; the closest institutional treatment, which we draw on directly instead, is Gobetti and Orair (2017)'s reconstruction of Brazil's above-the-line fiscal accounts net of the same operations (Section A.11).

A.11 Brazilian macro-fiscal evidence

Sanches (2026) update Brazilian federal fiscal multiplier estimates by spending and revenue instrument using a structural approach over federal accounts from 1997 to 2023, and Carvalho and Sanches (2019) is the antecedent study on which the present model's package-specific multipliers draw. Gruss (2014) estimates how strongly Latin American and Caribbean output responds to commodity terms-of-trade shocks and finds, in an input this model inherits directly, that Brazil's response is markedly smaller than its regional peers; our own back-test (the main manuscript's back-testing section) suggests this may understate the 2003–2010 commodity-boom experience once capital inflows and valuation effects are considered. Bom and Ligthart (2014) provide the meta-analytic range for the output elasticity of public capital that this model uses, pending the Brazilian public-capital stock data needed to activate that channel, and Ostry (Berg and Tsangarides) is the IMF staff discussion note on redistribution, inequality

and growth that motivates an inequality-to-growth feedback the model specifies but, pending a wired Gini path, does not yet activate.

Two further bodies of Brazil-specific evidence bear directly on this paper's own findings. First, on the fiscal reaction function Section A.3 flagged as the natural check on any debt-sustainability exercise: de Mello (2008) estimates a conventional, Bohn-style reaction function for Brazil across three levels of government (1995–2004) and finds the primary surplus responds positively and significantly to debt, in line with the typical emerging-market panel result; Mendonça (Santos and Sachsida) refine this with a Markov-switching specification and find a structural break around 2000, with a strong reaction pre-2000 and a weak-to-absent one thereafter. Both estimates are of the *conventional* sign, the opposite of the small, negative Brazilian-specific coefficient this paper's own D3 debt-panel review finds over 2000–2024 (Appendix B), a discrepancy we read not as a contradiction but as evidence that Brazil's fiscal reaction has itself shifted, plausibly again, since the mid-2000s sample these two papers cover, an open question this paper does not resolve. Second, on the stock-flow-adjustment residual at the centre of this paper's own back-test failure: Gobetti and Orair (2017) reconstruct Brazil's above-the-line fiscal accounts net of the same discretionary operations (Treasury-to-BNDES lending, Central Bank result transfers, and the broader *pedaladas fiscais* episode) that this paper's own residual isolates, the closest peer-reviewed treatment of that phenomenon available; Orair and Gobetti (2017) provide the companion narrative of Brazilian fiscal policy's swing from 2005–2010 expansion through 2011–2014 tax cuts to post-2015 austerity, useful context for interpreting why the residual is concentrated in that period. Bevilaqua and Garcia (2002) and Vivian and Mendonça (2008) cover Brazilian public-debt management strategy more broadly; Nunes et al. (2020) show how Central Bank profit transfers have themselves been used to help satisfy the constitutional golden rule even as public investment has lagged debt growth, a fiscal-rule-gaming concern directly relevant to Section A.9; and the Brazilian Senate's own Independent Fiscal Institution provides the most current technical evaluation of the 2023 spending framework modelled in the main manuscript's scenario-design section, flagging its reliance on uncertain, court-dependent revenue (Instituição Fiscal Independente, 2023), a companion technical note on integrating the golden rule and the spending ceiling into one coherent architecture (ENAP / Tesouro Nacional, 2019). Finally, Alston et al. (2013) frame Brazil's post-1985 fiscal and institutional trajectory as a slow change in beliefs about the social contract itself, a political-economy lens worth bearing in mind whenever two candidate programmes are described, as this paper describes them, as formally interchangeable closure rules for the same debt-accumulation identity.

B Diagnostic extensions: how much would a reaction, or a nonlinear premium, change the picture?

This appendix gives the full detail summarised in the main manuscript's summary of these diagnostics. Tier 2's failure (the main manuscript's back-testing section) leaves open a question the back-test itself cannot answer: how much of the headline status-quo debt path is an artefact of the model's deliberately simple, no-policy-reaction specification, and how much would a more behaviourally realistic government, or a more aggressively nonlinear market response, change it? Rather than fold a speculative answer into the headline results, we report it as a clearly labelled, off-by-default diagnostic, run only against the status-quo baseline (S0) itself, never differenced against or compared with the policy scenarios of the main manuscript's scenario-design section, and excluded from every published Monte Carlo summary. Three hundred draws (central draw plus 299 sampled, seed 20261020, an RNG stream independent of every other Monte Carlo channel) are used throughout. Every absolute debt figure in this appendix,

breach probabilities, debt levels, the fiscal gap, inherits the same Tier 2 bias documented in the main manuscript's back-testing section: the model's own seven-year recursion overshoots observed debt by 12.5–16.1 pp of GDP even after the best available correction, and nothing here adjusts for that overshoot. We flag this once, at the top, rather than repeat it after every number, but it applies with equal force throughout, including to the breach probabilities below, which should be read as upper bounds conditional on an uncorrected model rather than as calibrated risk estimates.

First, an EC/IFI-style probability-of-breach and fiscal-gap statement replaces the bare median debt path with a distribution. Table 1 reports the probability that S0's debt ratio exceeds 90%, 100% and 120% of GDP at each horizon, with and without the stochastic stock-flow-adjustment term of the main manuscript's back-testing section; Table 2 reports the primary balance that would stabilise debt at each horizon's own implied interest-growth differential (in the spirit of the European Commission's S1/S2 fiscal-gap indicators), against S0's actual projected primary balance. One feature of Table 1 is worth flagging rather than smoothing over: adding the (demeaned) stock-flow-adjustment term slightly *lowers* the five-year breach probability (78.6% to 73.6%) before the usual pattern of added uncertainty raising breach probabilities reasserts itself at longer horizons. We attribute this to the interaction between the bootstrapped adjustment and the debt-dependent risk-premium feedback channel of the main manuscript's model-architecture section, which can make a mean-preserving spread shift probability mass in either direction around a fixed threshold depending on where the threshold sits in the resulting distribution; we have not fully decomposed this interaction and report the counter-intuitive short-horizon result rather than suppress it.

Table 1: Probability that S0 debt/GDP exceeds stated thresholds, by horizon (300 draws).

Horizon	$P(>90\%)$, no SFA	$P(>90\%)$, SFA	$P(>100\%)$, SFA	$P(>120\%)$, SFA
5 years	78.6%	73.6%	35.8%	0.7%
10 years	97.7%	95.0%	86.3%	61.5%
15 years	100.0%	99.7%	98.0%	89.0%
20 years	100.0%	100.0%	100.0%	98.3%

Table 2: Stabilising primary balance and fiscal gap, S0, pp of GDP. Median [p10, p90] across 300 draws.

Horizon	Stabilising primary balance	Actual projected primary balance	Fiscal gap
5 years	$+2.60$ $[-0.22, +4.84]$	-2.39 $[-2.44, -2.22]$	$+4.85$ $[+2.11, +7.16]$
10 years	$+3.60$ $[-0.14, +7.16]$	-4.04 $[-4.17, -3.68]$	$+7.42$ $[+3.74, +11.13]$
20 years	$+7.95$ $[+0.65, +17.26]$	-7.68 $[-8.45, -6.80]$	$+15.63$ $[+8.06, +25.10]$

Second, we report two off-the-side variants of S0 itself, holding its own interest-growth path fixed and changing only the primary-balance rule or the risk-premium shape, so that each isolates a single mechanism. `frf_emerging` recomputes the primary balance each year as $pb_t = pb_{S0,t} + \rho\,(d_{t-1} - d_{2025})$ using the typical emerging-market panel's estimated long-run fiscal-reaction coefficient ($\rho = 0.0334$). Brazil's own estimated coefficient over 2000–2024, an in-house estimate produced for this project's own debt-panel review, not an independently peer-reviewed figure, and reported here with that provenance made explicit, is small and of the *opposite* sign (-0.037), in contrast to the conventional, positive, peer-reviewed coefficients de Mello (2008) and Mendonça (Santos and Sachsida) estimate for Brazil over earlier, pre-2005 samples (Section A.11); we therefore also report `frf_brazil_robustness`, which applies this in-house

coefficient instead, read as a robustness check on an internal estimate rather than as a published external finding. `premium_nonlinear` replaces the linear, capped risk-premium channel of the main manuscript's model-architecture section with the fiscal-fatigue-style shape theorised by Ghosh et al. (2013) and Bi (2012) (Section A.7): zero below a historical threshold debt level (119.6% of GDP, the 95th percentile of a fifty-country comparator panel) and rising with the square of the excess above it, capped at 2,000 basis points to avoid the numerical explosion an uncapped quadratic specification produces. Table 3 reports all three alongside plain S0, where "plain S0" is the *median across the 299 sampled draws*, not the single deterministic central-parameter projection reported in the main manuscript's model-architecture and headline-results sections (145.5% of GDP at this horizon). The two are different statistics and need not coincide: the median of a resampled, compounding debt-path distribution is not generally equal to the same path evaluated at the sampled parameters' own central values, since the debt identity compounds multiplicatively and the sampled distribution is right-skewed. We report both, rather than letting one stand in for the other, precisely because conflating them would overstate how tightly the two exercises agree.

Table 3: S0 debt/GDP (%) under diagnostic reaction and premium variants. Median [p10, p90] across 299 sampled draws throughout, including the "plain S0" column (distinct from the main manuscript's own 145.5%-at-ten-years deterministic central projection; see text).

Horizon	Plain S0 (sampled median)	`frf_emerging`	`frf_brazil_robustness`	`premium_nonlinear`
10 years	130.4	124.0 [99.3, 145.6]	138.9 [104.8, 168.8]	130.5 [101.7, 157.1]
20 years	245.5	206.7 [146.4, 270.9]	308.8 [184.8, 441.8]	487.8 [165.8, 1080.0]

Three readings of these tables deserve emphasis, each a direct application of the literature surveyed in Sections A.3–A.11. First, under the model's own preferred specification (including the stochastic stock-flow-adjustment term), the probability of breaching even a comparatively undemanding 90% of GDP threshold is already 73.6% at five years and 95.0% by ten (Table 1's "SFA" columns, not the more optimistic "no SFA" columns), a statement the main manuscript's headline-results table's paired differences cannot make on their own, and precisely the kind of risk statement Garcia and Rigobon (2004) and the growth-at-risk literature (Prasad et al., 2019; Lian, Presbitero and Wiriadinata; Furceri et al., 2025) argue a central path alone obscures. Second, the `frf_brazil_robustness` variant, using Brazil's own, historically estimated reaction coefficient rather than the emerging-market average, produces a *higher*, not lower, debt path than plain S0, the arithmetic consequence of that coefficient's counter-intuitive sign; we are careful to present this only as a robustness check on what Brazil's own history implies, following Simonassi and Lima (2005)'s related finding that Brazilian fiscal reaction may itself be regime-dependent, and explicitly not as a forecast that Brazilian fiscal policy will behave this way going forward. Third, the nonlinear premium variant is flat and close to the linear specification below the historical threshold but diverges sharply above it, reaching a 90th-percentile debt ratio of 1,080% of GDP at twenty years, a number we report not as a plausible outcome but as a crisis-territory flag in the sense Ghosh et al. (2013) and Bi (2012) intend: once a fiscal-fatigue-style premium is allowed to operate, paths that cross the historical threshold do not approach a new equilibrium, they explode, which is itself the theoretical point. None of the three variants alters any other published Monte Carlo output, and none is a forecast: each isolates a single mechanism against an otherwise unchanged baseline, in the same spirit as the channel-by-channel attribution already used for the long-run feedback block (the main manuscript's model-architecture section).

C Diagnostic extensions, part II: robustness checks on the business cycle, monetary policy, the exchange rate and interest-income incidence

A natural sanity check on the model's own completeness (summarised in the main manuscript's discussion section) is whether it is acyclical, monetary-policy-blind and currency-blind, and whether this quietly biases the finding that expansion scenarios lower the Gini coefficient (the main manuscript's headline-results table) in favour of expansion, because the model never prices the channels a sustained, debt-financed expansion would plausibly trigger: a Selic hike, a weaker real, and interest income accruing disproportionately to bondholders concentrated at the top of the income distribution. This appendix reports, in full, six off-by-default diagnostics built to test exactly this, each isolating one mechanism against the unchanged baseline, in the same spirit as Appendix B. None is folded into the main manuscript's headline-results table or any other published Monte Carlo summary; each uses 300 draws (central draw plus 299 sampled, seed 20261026, independent RNG streams) unless stated otherwise. Section C.5 adds a seventh, different in kind from the other six: not a Monte Carlo diagnostic at all, but a direct test of whether MR1's reactive-Selic mechanism, used as a stand-in for the model's own fixed-rate assumption, actually improves the fit to Brazil's observed fiscal history.

C.1 Does the model consider the business cycle?

This sanity check's starting diagnosis is correct, and independently confirmed twice over. First, this model's own deterministic growth trend has no cyclical fluctuation whatsoever: under the central draw with no external shocks, S0's year-on-year real GDP growth decelerates smoothly and monotonically from 1.18% in 2026 to 0.94% by 2035, with no booms or slowdowns at all. Second, a historical fiscal-timing diagnostic against Brazil's own 2000–2024 record (gross domestic product growth against its own trailing five-year mean, and the year-on-year change in the primary balance) finds fiscal policy was predominantly *countercyclical* over the period, not procyclical: 2009 and 2020–2021 are both clearly countercyclical, and only two years in the whole sample, 2006 and 2023, classify as procyclical-expansionary. The specific 2011–2014 period, sometimes singled out in qualitative accounts of this episode, is mixed rather than uniformly procyclical: 2011 is procyclical-contractionary, but 2012–2014 are all countercyclical on this measure, a finding that partly complicates, rather than confirms, the common qualitative reading of that episode.

Reintroducing genuine cyclical variation via the model's existing stochastic external-shock bootstrap (the main manuscript's model-architecture section) and rescaling the fiscal multiplier by a sampled recession/expansion ratio (low 1.0, central 1.5, high 2.5, reflecting Auerbach and Gorodnichenko (2012)'s qualitative finding of larger multipliers in recessions and Ramey (2019); Ramey and Zubairy (2018)'s more sceptical reading of the same literature; Pires (2014), cited elsewhere in this paper for Brazilian multiplier estimation generally, is not used for this specific ratio, since a secondary source suggests its own positive finding concerns a volatility-regime split rather than a business-cycle-phase split) moves R_AMB's median ten-year GDP cost from −0.80% under the model's fixed multiplier to −1.08% once state-dependence is reintroduced, roughly 35% larger, consistent with the standard state-dependent-multiplier intuition that ignoring state-dependence understates the true cost of a sustained contraction.

C.2 Debt composition and the exchange rate

The debt identity (the main manuscript's model-architecture section) applies one interest rate to an undifferentiated stock and revalues foreign-currency-linked debt only indirectly, through inflation pass-through, never as a direct stock revaluation. The main manuscript's discussion

section already reports why this channel is structurally small for Brazil specifically (a stable 3.7–4.4% foreign-exchange-linked share of federal debt, Secretaria do Tesouro Nacional, RMD); this appendix quantifies it. Table 4 revalues that FX-linked share directly by the model's own bootstrapped BRL/USD path (common to every scenario, as in the main manuscript's model-architecture section). The effect is indeed small at the paper's main ten-year horizon, as the main manuscript's discussion section's framing expects, but grows to a no-longer-negligible median 5–17 pp of GDP by twenty years, because the small FX-linked tranche is revalued by *cumulative* depreciation, and Brazil's own 2000–2024 BRL/USD history includes twenty-year windows with well over 100% cumulative depreciation; this remains modest relative to the already very wide uncertainty band on total debt/GDP at that horizon, but the "second order, do not worry about it" framing should not be extended to the twenty-year horizon without this qualification.

Table 4: FX-linked debt revaluation: additional debt/GDP (pp) from revaluing the FX-linked tranche by the common bootstrapped BRL/USD path. Median [p10, p90], 300 draws.

Scenario	10 years	20 years
S0	+3.21 [−0.48, +10.39]	+16.93 [+1.84, +47.50]
R_MOD	+2.92 [−0.43, +9.44]	+13.46 [+1.38, +37.54]
R_AMB	+2.63 [−0.39, +8.46]	+9.79 [+0.94, +28.46]

Separately, FX1 asks whether the exchange rate should itself respond to a scenario's own fiscal stress, rather than only to the common, policy-blind path above. Rather than fit a new debt-to-FX regression — this paper's own external-sector block already tried the closest analogue, a debt-to-sovereign-spread relationship for Brazil (fit 2000–2014, tested 2015–2024), and found it not supported out of sample — FX1 chains the model's own already-estimated debt-to-risk-premium parameter (Baldacci and Kumar, 2010) to a sampled, not fitted, spread-to-depreciation elasticity anchored on uncovered interest parity, with its upper bound informed only qualitatively by Banerjee (Mehrotra and Zampolli)'s finding that the fiscal-to-exchange-rate channel is markedly stronger in emerging than advanced economies. A first implementation routed this extra, stress-induced depreciation through the same inflation-pass-through channel as the common path above and found an *exactly zero* debt/GDP effect at every draw: in this model's Fisher-consistent debt identity, the nominal interest rate and nominal GDP growth both reprice by the same inflation factor, so a pure inflation shock with no real-economy effect cannot move the debt/GDP ratio at all, by construction — an independently confirming structural fact about the model, in the same spirit as the acyclical growth core reported above. FX1 therefore revalues the same FX-linked tranche as Table 4, but by the scenario's own stress-induced depreciation rather than the common path. Table 5 shows this is, on its own, a small effect for the paper's expansion scenarios, confirming that the FX-linked channel stays small specifically because the underlying FX-linked share of debt is small, not because fiscal stress fails to move the exchange rate.

Table 5: FX1: additional debt/GDP (pp) from a stress-induced, scenario-specific depreciation revaluing the FX-linked tranche. Median [p10, p90], 300 draws.

Scenario	10 years	20 years
E_MOD	+0.18 [+0.06, +0.36]	+0.42 [+0.16, +0.83]
E_AMB	+0.21 [+0.08, +0.40]	+0.47 [+0.18, +0.93]

C.3 Interest income and the inflation tax

The most direct distributional question this sanity check raises is whether higher government interest expense accrues disproportionately to bondholders at the top of the income distribution, and whether an inflation tax falls hardest on poor households' non-indexed transfers — neither of which the household microsimulation (the main manuscript's model-architecture section) had any mechanism to represent, since the underlying survey has no capital-income field. IY1 derives a scenario's change in government interest expense directly from the debt identity (the previous year's debt/GDP times that year's nominal interest rate), allocates it across the simulated population using the concentration of declared capital and financial income among Receita Federal's own tax centiles (Receita Federal, 2024) (the top 1% of declarants hold 68.3% of declared capital/financial income; the top 10% hold 90.7%), and separately erodes each household's baseline Bolsa Família amount by that scenario-year's cumulative extra inflation. Table 6 reports the resulting shift in the Gini coefficient against the published, IY1-off figure, for the two debt-financed expansion scenarios this check is specifically about.

Table 6: IY1: shift in the Gini coefficient (points) from switching on the interest-income and inflation-tax channels, against the published figure. Median [p10, p90], 300 draws.

Scenario	10 years	20 years
E_MOD	+0.76 [+0.41, +1.16]	+3.13 [+1.60, +4.56]
E_AMB	+1.31 [+0.72, +1.98]	+5.02 [+2.48, +7.19]

Poverty (the R$218 line) barely moves anywhere in this table (well under one percentage point throughout), which confirms the capital-income windfall, not the inflation-tax erosion of Bolsa Família, is the dominant channel here: poverty is measured at the bottom of the distribution, and the windfall is concentrated at the top. The direction is exactly what a bondholder-concentration channel would predict, and the magnitude is not negligible against the published figure it is being compared with, which we turn to directly in Section C.6 below.

C.4 A reactive Selic

A further sanity check concerns whether the model has any monetary-policy reaction function at all: the real interest rate is a fixed input regardless of inflation or the output gap a scenario itself creates. Rather than make this shared rate-determination function itself state-dependent, which would touch every scenario, every other diagnostic in this appendix, and the published Monte Carlo at once, MR1 re-runs each scenario's debt path afterwards, substituting a textbook Taylor rule (Taylor, 1993) for the fixed rate in an otherwise identical re-run of the debt identity. Because this model's inflation input does not vary by scenario, only the output-gap term differentiates a scenario's Selic path from S0's. The Brazil-specific Taylor-rule literature we could locate (several studies spanning 2001–2025) consistently reports a positive, significant Selic response to both the inflation and output gap, but only qualitatively ("smaller than advanced economies," "varying over time"); rather than use an unverified point estimate, the coefficient is sampled as a discount (low 0.3, central 0.6, high 1.0) on Taylor's own classic 0.5 coefficient. Table 7 reports the resulting incremental debt/GDP, on top of the already-reported fixed-rate baseline differential (the main manuscript's headline-results table).

C.5 Testing MR1 against Brazil's own fiscal history

MR1 above answers one question: within a Monte Carlo draw, how would a reactive Selic change a scenario's debt path relative to S0? A different and harder question is whether

Table 7: MR1: incremental debt/GDP (pp) from a reactive Selic, on top of the fixed-rate baseline differential vs. S0. Median [p10, p90], 300 draws.

Scenario	10 years	20 years
E_MOD	+1.46 [+0.83, +2.39]	+5.45 [+2.92, +10.26]
E_AMB	+3.02 [+1.71, +4.95]	+6.32 [+3.28, +12.66]

MR1's mechanism, used not as an add-on to a scenario comparison but as a direct replacement for the model's own fixed real-rate assumption, actually describes Brazil's observed 2000–2024 fiscal history better than the fixed rate it would replace. This is the same distinction the main manuscript's back-testing section already draws for the published model's own back-test, Tier 1 (differences) against Tier 2 (absolute levels): a mechanism can be defensible for one purpose and not the other, and the two tests need not agree.

We built a separate historical reconstruction, outside any Monte Carlo draw: starting from Brazil's own observed general-government debt/GDP in 2006, the debt identity is recursed annually through 2024 using the actually observed primary balance and real GDP growth every year, but *generating*, not observing, the interest rate under two candidate mechanisms — the model's existing fixed real rate (5.05%, itself the historical median real rate the model already uses, not an arbitrary flat guess) and MR1's Taylor rule, reacting each year to that year's own observed inflation and a real-time output-gap proxy identical to the one Appendix B's PCY1 diagnostic already uses. Two debt-level tests compare each mechanism's resulting path to the actually observed one: a cumulative reconstruction from the single 2006 anchor, which the main manuscript's back-testing section already documents as dominated by stock-flow-adjustment noise unrelated to any rate mechanism, and a one-step reconstruction re-anchored on the observed lagged debt every year, which isolates rate-mechanism fit from that noise and is accordingly the test we treat as decisive. A third test compares each mechanism's own implied nominal rate directly to the actually observed Selic.

Table 8 reports the result, confirmed against two independently constructed nominal-growth series (a CPI-and-real-growth approximation, and Brazil's own SIDRA current-price GDP series, used elsewhere in this paper's growth-block back-test). The two tests disagree, and we report that disagreement plainly rather than resolve it in either direction: the Taylor rule's own implied rate tracks the actually observed Selic more closely than the fixed-rate assumption does, but feeding that same generated rate through the debt identity produces a *worse* one-step reconstruction of the observed debt path than the fixed-rate assumption. We read this as a category error our own design would have invited, not a contradiction: the fixed rate is calibrated to the government's effective, blended rate across its whole diversified debt stock (Selic-linked, inflation-linked and fixed-rate instruments alike, the main manuscript's discussion section), not the Selic policy rate itself, and a rule built to track Selic is accordingly the wrong direct stand-in for it. We therefore do not adopt MR1's mechanism as a replacement for the fixed-rate assumption, in the published Monte Carlo or elsewhere; Table 7's own incremental figures remain exactly what they already were, a Tier-1-style scenario-difference diagnostic, not a claim that the underlying mechanism is a validated improvement on the model's historical fit. Nominal GDP growth in the CPI-based series is a Fisher approximation (real growth compounded with CPI inflation) rather than a directly observed nominal series; this affects both mechanisms' reconstructions equally and so does not bias the comparison between them, which is the comparison this test exists to make. A wider benchmark of this project's own validation practice against the literature (Appendix D) found the European Stability Mechanism's own operational fan-chart methodology reports a calibrated coverage statistic rather than a single pass/fail verdict; applied here, at no new estimation cost, using the model's own already-

published low/high real-rate corners to bound the one-step reconstruction rather than a single central line, the actually observed debt/GDP fell inside that band in 9 of the 14 reconstructed years (64%), missing in 2011, 2012, 2014, 2015 and 2022 — a second, complementary way of reading the same Tier-2 shortfall the main manuscript's back-testing section already reports, not a contradiction of it.

Table 8: MR1 against history: one-step debt/GDP reconstruction error and nominal-Selic tracking error, fixed-rate vs. Taylor-rule mechanism, 2000–2024 (RMSE, percentage points).

Mechanism	Debt/GDP, one-step (CPI-based)	Debt/GDP, one-step (SIDRA-based)	Nominal Selic
Fixed rate	3.69	3.23	3.87
Taylor rule	4.25	3.88	3.35

C.6 Putting it together: does the omitted-channel bias hold up?

The hypothesis under test is that the Gini and debt findings of the main manuscript's headline-results table are probably biased in favour of expansion. Combining each diagnostic's own median effect with the corresponding published figure — a simple addition of two separately-run statistics, not a single re-integrated Monte Carlo, since medians of compounding, nonlinear quantities do not generally add linearly, so this is indicative rather than a recomputed joint distribution — gives a direct answer for the two channels large enough to matter at the paper's main ten-year horizon.

On distribution: the published finding is that E_MOD lowers the Gini coefficient by a median 0.10 points and E_AMB by 0.16 points at ten years (the main manuscript's headline-results table). Adding IY1's own median shift (Table 6) gives a combined $-0.10+0.76 = +0.65$ points for E_MOD and $-0.16 + 1.31 = +1.15$ points for E_AMB: the median effect does not merely shrink, it *reverses sign*. On this reading, the hypothesis is not just plausible but directionally confirmed: once interest income is distributed the way Receita Federal's own data says capital income actually is, the model's central estimate is that these expansions make the Gini coefficient worse, not better, at the median. On debt: adding MR1's and FX1's own median effects (Tables 7 and 5) to the published differential raises E_MOD's ten-year debt cost from +11.6 to roughly +13.3 pp of GDP and E_AMB's from +23.3 to roughly +26.5 pp, about 13–14% larger in each case — a real, quantified addition, though one that changes the *size* of the debt cost rather than its sign.

We think this is the honest answer our own stated methodological standard demands, stated plainly rather than hedged: the distributional claim this paper previously reported without these channels does not survive their inclusion at the median, and the debt claim, while directionally unchanged, is understated by a magnitude worth citing rather than waving away. We did not set out to find this; PCY1 and FX1, by contrast, *complicate* rather than confirm parts of this hypothesis (fiscal policy's own history is predominantly countercyclical, and the FX-linked debt channel is genuinely small once isolated from the common external path), and CYC1's and DC1's findings are each real but, at the ten-year horizon that is this paper's main reference point, modest. The net reading is neither "the hypothesis is simply right" nor "the original paper's finding stands": it is that the specific channel we pressed hardest on testing, interest income flowing disproportionately to the already-wealthy, is large enough to reverse a headline distributional finding once it is actually quantified, and that is precisely the kind of result an honest accounting exercise exists to surface, whichever way it comes out.

D Diagnostic extensions, part III: a wider methodology sanity check

A self-administered benchmark of this project's own debt-sustainability methodology against the wider literature, undertaken before submission, surfaced two further sanity checks worth running directly against this model, reported here in the same off-by-default, never-folded-into-the-headline-results spirit as Appendices B and C.

D.1 Does joint shock correlation widen the model's own uncertainty bands?

The published Monte Carlo samples its structural parameters (growth, the interest rate, multipliers) independently per draw. Two recent institutional debt-sustainability exercises outside this project — a Bayesian and frequentist VAR analysis of France's debt trajectory (Baret, Bec and Cochard) and the European Stability Mechanism's own operational euro-area fan-chart methodology (Hitaj, Zavalloni and Zigraiova) — instead draw growth, inflation, the interest rate and the primary balance as a single, jointly-correlated shock, and both report this produces *wider*, not narrower, uncertainty bands than independent or frequentist-only sampling. We test the same question against Brazil's own 2002–2024 history directly: a joint three-year block bootstrap of growth, inflation, the primary balance and the exchange rate together (preserving whatever correlation they actually had in a given historical window, the same block-bootstrap convention, at the same block length, already used for this paper's own stock-flow-adjustment term in the main manuscript's back-testing section), compared against the published Monte Carlo's own band width at the same scenario and horizon. A fitted vector autoregression was deliberately not used: the overlapping annual window for all four series is only 23 years, too thin to estimate a multi-variable VAR with any confidence distinguishable from noise.

Table 9 reports the result. The correlated-shock band is wider than the published Monte Carlo's own band in 16 of 20 scenario-horizon cells checked, by a wide margin at longer horizons — roughly double the published band's width at twenty years for the ambitious scenarios. Brazil's own historical experience is consistent with the same pattern found elsewhere: treating growth, inflation, the primary balance and the exchange rate as independently drawn, as the published Monte Carlo currently does, plausibly understates the true joint tail risk around the headline debt path. We do not adopt this correlated-shock structure as a replacement for the published sampling scheme — that would be a larger redesign than a single diagnostic justifies — but report it as evidence the published bands are, if anything, too narrow rather than too wide.

Table 9: Band width (p90−p10 of debt/GDP vs. S0, pp): published Monte Carlo vs. the jointly-correlated-shock diagnostic, 300 draws.

	10 years		20 years	
Scenario	Published	Correlated	Published	Correlated
R_MOD	1.69	2.86	17.21	36.08
R_AMB	3.38	5.73	34.77	72.69
E_MOD	1.69	2.87	17.22	36.09
E_AMB	3.38	5.74	33.60	69.71

D.2 A climate/natural-disaster tail-shock channel

The debt-sustainability literature has, in the last two years, institutionalised climate and natural-disaster shocks into standard debt-sustainability toolkits, calibrating a tail shock to

real growth and the primary balance together at a severe historical percentile (Draganic, Srdelic and Davila-Fernandez). This model already has a structurally close analogue for a different tail risk: the governance layer's major-corruption-shock process (the main manuscript's model-architecture section), a stochastic annual event with a triangularly-distributed magnitude and a build-up/decay GDP-level path, applied identically to every scenario within a draw. We reuse that exact machinery for a climate/disaster shock instead of building a new one, adding the one genuine extension the comparator toolkits call for: a shared-draw, direct primary-balance hit (emergency relief spending), since a disaster shock is not GDP-level-only the way the corruption shock is.

No general-purpose, Brazil-specific disaster-fiscal-cost dataset was identified; the standard international disaster database is independently documented to understate Brazilian disaster losses by roughly an order of magnitude relative to Brazil's own national disaster-registry system. Rather than adopt an uncalibrated cross-country figure, the magnitude is anchored on a single, large, well-documented federal-level event: the 2024 Rio Grande do Sul floods, whose federal fiscal cost is reported at R$7.7–13.4 billion, approximately 0.07–0.12% of GDP. This is one severe event, not an estimated distribution, and the sampled low/central/high range brackets it with a proportionate, judgemental band, in the same spirit already applied elsewhere in this paper to a single-estimate-anchored coefficient (CYC1's state-dependence ratio, FX1's depreciation elasticity).

Table 10 reports the resulting additional debt/GDP. The median effect is small at every horizon checked (under a quarter of a percentage point of GDP at twenty years for every scenario), but the right tail is not: the 90th percentile reaches 2–4.5 pp of GDP by twenty years, a genuine, if low-probability, addition to the uncertainty this model's published bands do not currently carry.

Table 10: Additional debt/GDP (pp) from the climate/disaster-shock diagnostic, median [p10, p90], 300 draws.

Scenario	10 years	20 years
S0	+0.12 [0.00, +1.58]	+0.15 [0.00, +3.43]
R_MOD	+0.11 [0.00, +1.44]	+0.11 [0.00, +2.76]
R_AMB	+0.10 [0.00, +1.29]	+0.08 [0.00, +2.19]
E_MOD	+0.13 [0.00, +1.72]	+0.18 [0.00, +4.05]
E_AMB	+0.14 [0.00, +1.86]	+0.21 [0.00, +4.50]

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